

Online Appendix for

The Price of Exclusion: SME Set-Asides in Public Procurement

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This appendix reports the checks behind the identification and policy claims in the paper: institutional timing, reduced-form validation, structural cost recovery, mechanism robustness, coordination screens, and welfare calculations. The main text keeps only the evidence needed for the argument. The appendix records the diagnostics that support, qualify, or bound that interpretation. Appendix OA-A documents the institutional setting; OA-B reports reduced-form diagnostics; OA-C covers structural cost recovery; OA-D reports mechanism and coordination robustness; and OA-E gives welfare and policy-design details.

A Institutional Details

A.1 Legal timeline

Brazil’s federal SME procurement framework begins with LC 123/2006, which defined the SME categories used in procurement policy. The statute distinguishes MicroEmpresas, with annual revenue up to R\$360,000, from Empresas de Pequeno Porte, with annual revenue up to R\$4,800,000 after the LC 155/2016 update. LC 147/2014 made SME-only procurement mandatory for eligible purchases below R\$80,000, but implementation depended on how procurement authorities interpreted the threshold.

In São Paulo, the relevant interpretation changed from a purchase-notice view to an item-by-item view. Under the restrictive view, a purchase notice whose total value exceeded the statutory threshold was not treated as SME-only even when individual items were below the threshold. The item-level interpretation made many more items eligible for SME-only treatment. BEC communicated the technical implementation through Comunicados 02/2017 and 03/2017, while operational take-up in Group 65 occurs around March 2018. I therefore treat March 2018 as the empirical cutoff: it is the mass-adoption date, not the first legal moment at which SME preferences existed.

A.2 Platform, products, and sample

The empirical setting is the BEC electronic procurement platform, created under Decrees 45.085/2000 and 45.695/2001 and later used for Pregão auctions under Decree

Table OA-1: Institutional timeline

Instrument	Date	Role in the empirical design
LC 123/2006	2006	Federal SME statute; defines SME procurement treatment and preference tools.
13.122/2008	2008	São Paulo state SME procurement law.
LC 147/2014	2014	Makes SME-only procurement mandatory for purchases below R\$80,000.
TCE-SP eTC 5509.989.15	2015	Illustrates the restrictive interpretation of the threshold.
PGE-SP Parecer 151/2017	2017	Supports the item-level interpretation used in BEC implementation.
BEC Comunicados 02/2017–03/2017	2017	Platform communication enabling SME-only and mixed-exclusivity functionality.
March 2018	2018	Mass take-up of SME-only treatment in Group 65; empirical cutoff.
TCE-SP eTC 9589.989.18	2018	Later ruling consistent with the expansive item-level interpretation.

The table separates legal authority from empirical timing. The design uses the observed mass take-up in Group 65 rather than treating any single legal document as the treatment date.

49.722/2005. The extract covers 36 months, from September 2016 through August 2019, and contains 4.8 million item-level transactions and 860,000 purchase offers. Group 65 accounts for 27 percent of platform transactions in this window, which is why the item-level reinterpretation has substantial empirical bite there.

The structural sample restricts attention to Pregão auctions with at least 2 bidders and normalized costs below $c_\epsilon = 3$. The upper-cost filter removes only 0.4 percent of bids, so the main results are not driven by discarding a large fraction of the auction data. The main sample contains 48,740 pre-period auctions and 49,253 post-period auctions; pharmaceutical Pregão auctions account for 59,997 auctions.

B Reduced-Form Evidence

B.1 Balance and scope of the comparison

The reduced form is a timing, sign, and scale check for the structural exercise, not the source of the decomposition. This distinction matters because Group 65 is an exposed product group rather than a randomized treated unit. In pre-period balance checks, Group 65 contributes 30718 items and the pooled controls contribute 92837 items across 77 never-treated groups. The largest absolute standardized difference is 0.599; 7 of 9 variables exceed 0.10 in absolute value and 4 exceed 0.25. These diagnostics justify the

main text’s conservative use of the DiD estimates. Table OA-2 reports the covariates behind this summary.

Table OA-2: Pre-treatment covariate balance: Group 65 versus never-treated controls

Variable	Group 65	Control-group mean distribution			Std. diff.	
	Mean	P25	Median	P75	Mean	Imbens–Rubin
Log reference price	4.036	4.057	5.385	6.874	5.657	-0.219
Bids per item	7.396	8.160	9.982	13.748	11.486	-0.176
Firms per item	2.784	3.203	3.681	4.347	3.765	-0.599
Share Pregão	0.645	0.358	0.471	0.805	0.559	0.445
Share items below R\$80k	0.966	0.886	0.964	0.992	0.907	-0.037
Share SME bids	0.000	0.021	0.075	0.104	0.068	-0.567
Share São Paulo-state bidders	0.704	0.719	0.794	0.835	0.768	-0.331
Log bidder-buyer distance	4.126	3.579	3.858	4.197	3.853	0.160
Final/reference price	0.683	0.595	0.626	0.661	0.641	0.010

Sample: pre-period items in the DiD panel, restricted to auctions with at least two bidders. Group 65 contributes 30718 items; the 77 never-treated control groups contribute 92837 items. Standardized differences are Imbens–Rubin normalized differences (Imbens and Rubin, 2015). The table is not a parallel-trends test; it documents cross-group composition that the fixed-effects design must absorb.

The imbalances in Table OA-2 are taken seriously. They do not by themselves test the identifying assumption. The relevant question is whether Group 65 had differential pre-trends or anticipatory breaks relative to the comparison groups, which is why the placebo and alternative-DiD diagnostics below are reported explicitly.

Table OA-3: Reduced-form diagnostics with reported standard errors

Diagnostic	Estimate	Standard error	Sample / notes
TWFE, headline estimator	-0.1080	0.0073	Item-clustered; 18-month window
BJS imputation estimator	-0.0560	0.0293	Robust imputation; post-treatment ATT
Callaway–Sant’Anna estimator	-0.0165	0.0932	Never-treated controls; simple post ATT
Phased-adoption check, full sample	-0.109	0.012	Platform enablement period retained
Phased-adoption check, excluding enablement months	-0.087	0.012	Months 690–697 excluded

Estimates use the paper’s sign convention. Negative coefficients mean the treated Group 65 price/reference discount narrows after SME-only adoption, so prices rise relative to the pre-policy benchmark. The headline TWFE row uses item fixed effects and item-clustered standard errors. The modern DiD rows are reported because they discipline the reduced-form validation rather than because the paper relies on their magnitude for the structural decomposition.

B.2 Placebo timing and alternative estimators

Placebo cutoffs before the reform are smaller in magnitude than the real-cutoff estimates: -0.013, -0.030, and -0.034. They are not zero in every case, so they do not support a

Table OA-4: Placebo timing checks on log prices

Fake cutoff	Estimate	Standard error	Observations
September 2017	-0.013	0.0182	311,883
March 2017	-0.030	0.0132	215,611
June 2017	-0.034	0.0128	237,625
Real-cutoff benchmark range	-0.108 to -0.142	—	—

Placebo regressions use pre-treatment data only, item and month fixed effects, sealed-bid and log-quantity controls, and item-level clustering. The placebo estimates are not all zero, so they limit the force of a reduced-form-only design. Their magnitudes are smaller than the real-cutoff benchmark, which is why the main text uses the DiD as validation for timing and sign rather than as the source of the structural price effect.

strong reduced-form-only interpretation. They do support the narrower claim used in the main text: the largest movement occurs around the institutional take-up date and has the sign predicted by the set-aside.

Modern DiD estimators reduce precision and attenuate the point estimate. The BJS imputation estimate is -0.0560 and the Callaway–Sant’Anna estimate is -0.0165, compared with a panel-aggregated TWFE estimate of -0.1080. The reduced form is therefore not asked to carry the structural price magnitude or a reduced-form welfare calculation. The decomposition uses the auction model and treats the DiD as external validation for timing and sign.

C Cost Recovery and Structural Estimation

C.1 Drop-out bids and winner censoring

In a Pregão reverse auction, losing firms exit as the price clock falls. Under the maintained IPV clock interpretation, those exits reveal willingness to supply at the auction scale. The winner is censored because the winning firm does not need to reveal the lowest price at which it would still supply. The baseline estimator treats the winning price as a tight upper-bound observation; the appendix checks a losers-only estimator and a Turnbull NPMLE (Turnbull, 1976) that treats the winning observation as left-censored.

The rejections in Table OA-7 are not treated as failed robustness checks. They are the empirical reason for using the observed post-policy SME pool in the baseline and reporting strict primitive invariance as a bounding exercise. The policy ranking is therefore emphasized where it survives that choice, especially in non-pharmaceutical markets.

Table OA-5: Winner-censoring and cost-recovery checks

Estimator	Net price effect		Deviation from baseline	
	Non-pharma	Pharma	Non-pharma	Pharma
All bidders, baseline	0.259	0.308	–	–
Losers only	0.275	0.347	6%	13%
Turnbull NPMLE	0.246	0.357	5%	16%

Net price effects are $p_{S_3} - p_{S_1}$, normalized by the buyer reference price. The largest Turnbull deviation from the baseline is 16 percent; in non-pharmaceuticals it is 5 percent.

The conclusion is not that censoring is irrelevant. It is that the central price-formation ranking does not hinge on the baseline treatment of the winner. The Turnbull estimator still gives exclusion shares of 74.0 percent in non-pharmaceuticals and 82.0 percent in pharmaceuticals.

Table OA-6: Cost-distribution quantiles under alternative winner treatments

Class	Type	Period	Median cost $c_{0.5}$			Upper-quartile cost $c_{0.75}$		
			Losers	All	Turnbull	Losers	All	Turnbull
Non-pharma	SME	Post	0.922	0.861	0.840	1.215	1.097	1.117
Non-pharma	SME	Pre	0.947	0.881	0.862	1.314	1.193	1.212
Non-pharma	Non-SME	Post	0.802	0.762	0.726	1.068	0.964	0.971
Non-pharma	Non-SME	Pre	0.768	0.719	0.684	1.028	0.918	0.932
Pharma	SME	Post	0.860	0.799	0.781	1.078	0.966	1.002
Pharma	SME	Pre	0.863	0.800	0.710	1.441	1.188	1.245
Pharma	Non-SME	Post	0.674	0.657	0.633	0.814	0.774	0.765
Pharma	Non-SME	Pre	0.619	0.604	0.574	0.762	0.725	0.708

Losers-only uses only drop-out bids and is upward-biased because it omits the lowest-cost bidder in each auction. All-bidders treats the winner’s final price as a point observation and is downward-biased if the final price exceeds true cost. Turnbull NPMLE treats the winner as left-censored at the final price.

C.2 Auction-level heterogeneity and cross-modality comparison

The baseline removes auction-level scale heterogeneity before simulating counterfactual bidder pools. The Pregão intraclass correlations range from 0.36 for non-pharmaceutical SMEs to 0.59 for pharmaceutical non-SMEs. These magnitudes are comparable to settings where auction-level unobserved heterogeneity is important: Krasnokutskaya (2011) reports an ICC benchmark of 0.66 in highway procurement.

The cross-modality check compares Pregão drop-out recovery with a GPV first-price recovery from Convite auctions (Guerre et al., 2000). The operational Kolmogorov–

Smirnov distance threshold is 0.05. The cleanest falsification cell, pharmaceutical non-SMEs in Convite, has distance 0.032. Non-pharmaceutical Pregão is close to the threshold (0.0486), while pharmaceutical Pregão before the heterogeneity correction is farther away (0.0722). This pattern supports the maintained correction but also motivates the main text’s cautious treatment of IPV.

Table OA-7: Primitive-stability and cross-modality diagnostics

Diagnostic cell	KS distance	Threshold	Interpretation
Non-pharma Pregão, raw drop-outs	0.0486	0.05	Close to the operational threshold.
Pharma Pregão, raw drop-outs	0.0722	0.05	Fails the threshold before common-scale cleaning.
Non-pharma Pregão, UH-clean	0.0613	0.05	Moves above the threshold after removing common scale.
Pharma Pregão, UH-clean	0.1407	0.05	Remains far from strict primitive invariance.
Non-pharma Convite GPV	0.0514	0.05	Borderline cross-modality comparison.
Pharma Convite GPV, non-SME	0.032	0.05	Cleanest validation cell; passes the threshold.
Convite pharma non-SME, post-UH	0.0225	0.05	Continues to pass after the heterogeneity correction.

The table reports Kolmogorov–Smirnov distances for diagnostics used to discipline the cost-recovery step. The goal is not to prove primitive invariance in every cell; the pharmaceutical Pregão cells visibly reject that strong restriction. The point is to locate where the recovery is most credible and where the paper must treat policy rankings as conditional.

D Robustness to the Core Mechanism

D.1 Strict invariance and protected-pool composition

The decomposition separates the effect of lost non-SME competition from the protected-pool offset. The weakest interpretive link is the offset, because $S_3 - S_2$ combines additional SME participation with changes in the active SME cost distribution. The strict-invariance benchmark therefore imposes the pre-policy SME cost distribution on the post-policy SME pool.

The non-pharmaceutical result is stable: the exclusion component remains the dominant force under baseline, strict invariance, Turnbull censoring, and alternative windows. The pharmaceutical result is informative but more conditional. Post-policy SME turnover is larger in pharmaceuticals: 61.9 percent of post-policy SME firms are new relative to the pre-period pool and these firms account for 37.8 percent of post-policy SME bids. In non-pharmaceuticals, new firms account for 23.0 percent of post-policy SME bids. Pharmaceuticals are therefore a boundary case for policy ranking rather than the headline market.

Table OA-8: Mechanism robustness

Specification	Net price effect		Exclusion share	
	Non-pharma	Pharma	Non-pharma	Pharma
Baseline	0.227	0.309	72.0%	68.8%
Strict invariance	0.29	0.47	85%	79%
Turnbull cost recovery	0.246	0.357	74.0%	82.0%
18-month window	0.272	0.292	79.7%	67.0%
12-month window	0.267	0.310	75.4%	73.0%
6-month window	0.269	0.338	76.4%	73.6%

Net effects are normalized by the reference price. Exclusion share is the absolute share of the decomposition attributed to removing non-SMEs from the price-forming pool. Rows combine different diagnostic families: strict invariance disciplines the protected-pool composition margin, Turnbull disciplines winner censoring, and window rows discipline timing sensitivity.

D.2 Entry counts, filters, and dependence

The baseline summarizes entry using class-by-period arrival rates. Window checks show little drift in the non-pharmaceutical exclusion share: 4.3 percentage points across the 18-, 12-, and 6-month windows. The pharmaceutical drift is larger, at 6.6 points, again matching the interpretation that pharma is composition-sensitive.

Cost-filter checks also preserve the qualitative result. Tightening the filter reduces the non-pharmaceutical net effect by 15 percent and the pharmaceutical net effect by 8 percent; loosening it raises the effects by at most 1 percent and 6 percent, respectively. Allowing within-auction cost correlation up to $\rho_c = 0.3$ moves the exclusion share by at most 5 percentage points and the total effect by at most 10 percent.

Table OA-9: Filter, window, and dependence sensitivities

Sensitivity	Non-pharma	Pharma	Interpretation
Tight cost filter: net effect	0.215	0.303	Smaller net effect after removing high normalized costs.
Baseline filter: net effect	0.254	0.330	Benchmark filter used in the robustness registry.
Tight-filter drop	15%	8%	Attenuation relative to the benchmark filter.
Loose-filter increase	1%	6%	Maximum increase under the looser filter.
Exclusion-share drift across windows	4.3 pp	6.6 pp	Difference across 18-, 12-, and 6-month windows.
Dependence grid	$\rho_c \in \{0, 0.1, 0.2, 0.3\}$		Total-effect drift at most 10%; exclusion-share drift at most 5 pp.

Net effects are $p_{S_3} - p_{S_1}$, normalized by the reference price. The dependence sensitivity allows within-auction cost correlation up to $\rho_c = 0.3$. The table does not require every specification to have the same level. It asks whether the exclusion-dominant decomposition survives the main filter, window, and dependence perturbations.

D.3 Coordination screens

Coordination is a first-order threat because repeated bidder pairs could make drop-out prices reflect conduct rather than costs. The screens detect clustering, but the important question for this paper is whether clustering intensifies after non-SMEs are removed. The Conley-style screen is stable in non-pharmaceuticals: realized shares are 16.9 percent before and 16.8 percent after the cutoff, compared with null means of 10.6 and 10.2. In pharmaceuticals, the realized share falls from 27.6 to 24.4 percent. Bajari–Ye-style ratios show the same directional pattern: the non-pharmaceutical ratio falls from 2.63 to 1.83, and the pharmaceutical ratio falls from 1.29 to 1.11.

These screens do not prove competitive conduct. They answer the narrower threat to the decomposition: the main post-policy price increase is not obviously driven by a new coordination shock among the surviving SME pool.

Table OA-10: Bidder-pair coordination screens

Stratum	Conley realized	Conley null mean	Bajari–Ye T_1 obs/null
Non-pharma Pre	16.9%	10.6%	2.63
Non-pharma Post	16.8%	10.2%	1.83
Pharma Pre	27.6%	16.3%	1.29
Pharma Post	24.4%	14.4%	1.11

The Conley-style statistic is the share of auctions in which the closest pair of UH-clean residual bids lies within a narrow reference-price band. The null re-samples bid sequences within cell while preserving bidder counts. The Bajari–Ye statistic is the ratio of the observed maximum pair co-bidding count to its permutation-null mean among repeated bidder pairs. Both screens detect baseline clustering. The identifying question here is the pre/post direction: neither statistic rises after the set-aside expands.

D.4 Reference-price validation

All structural prices are normalized by buyer reference prices, so mechanical changes in reference-price setting would be a problem. I therefore work with normalized cost and price objects and use the reduced-form price/reference benchmark as an external check. The structural net effects, 34 percent of the open-regime price in non-pharmaceuticals and 47 percent in pharmaceuticals, are larger than the DiD-implied benchmark of 12 percent. That gap is consistent with the fact that the structural exercise holds the relevant bidder pools fixed and simulates the order statistic directly, while the reduced form averages over take-up, controls, and product-time variation. The gap is a reason to interpret levels carefully, not a reason to discard the mechanism.

E Welfare Details

E.1 Lambda grid

The main text reports welfare losses at $\lambda = 0.30$. The welfare loss is

$$L(\lambda) = DWL^{alloc} + \lambda \Delta G,$$

where DWL^{alloc} is the allocative loss and ΔG is the extra government outlay. The grid ranges from 0.15 to 0.45 in increments of 0.10. Moving λ by 0.10 changes the welfare loss by about 3 percentage points in non-pharmaceuticals and 5 points in pharmaceuticals.

Table OA-11: Welfare loss across marginal-cost-of-public-funds values

Class	$\lambda = 0.15$	$\lambda = 0.20$	$\lambda = 0.30$	$\lambda = 0.40$	$\lambda = 0.45$
Non-pharma	24.1%	25.7%	28.9%	32.2%	33.8%
Pharma	38.1%	40.3%	44.8%	49.3%	51.6%

Losses are reported as percentages of the open-regime simulated price.

E.2 Preference grid and welfare weights

The 10 percent SME price preference is a benchmark instrument. It keeps non-SMEs inside the auction and tilts allocation through the scoring rule. In the simulated preference grid from 0 to 30 percent, the 10 percent preference is the cheapest welfare-dominant preference threshold in the current grid. At 10 percent, the price shift is -0.0036 in non-pharmaceuticals and +0.0015 in pharmaceuticals, while SME win rates rise by 4.3 and 1.4 percentage points.

The preference is not a costless substitute for the set-aside. At 30 percent, the welfare loss remains 4.99 percent in non-pharmaceuticals and 1.92 percent in pharmaceuticals, but the instrument still delivers less redistribution than full exclusion. The relevant comparison is therefore a frontier: how much additional SME surplus, access, or dynamic capacity does the planner require to justify removing non-SMEs from the price-forming pool? In the main specification, the implicit SME welfare weights required to prefer the set-aside to the 10 percent preference are 2.42 in non-pharmaceuticals and 2.61 in pharmaceuticals. Under strict invariance, the non-pharmaceutical comparison continues to require a weight above one, while the pharmaceutical comparison falls to 0.7.

Table OA-12: Preference grid: price effects, SME win rates, and welfare loss

Class	Preference	$\bar{p}$	Δ vs. S_1	% of V0 Δ	SME win rate	Welfare loss
Non-pharma	0%	0.772	+0.0046	2.0	17.1%	0.14%
Non-pharma	5%	0.768	+0.0015	0.7	20.0%	0.00%
Non-pharma	10%	0.764	-0.0028	-1.2	21.4%	0.25%
Non-pharma	15%	0.766	-0.0013	-0.6	24.6%	0.00%
Non-pharma	20%	0.775	+0.0081	3.5	27.8%	1.40%
Non-pharma	25%	0.790	+0.0227	9.7	30.2%	2.43%
Non-pharma	30%	0.809	+0.0420	17.9	34.0%	4.99%
Non-pharma	Full set-aside	1.001	+0.2342	100.0	100.0%	27.13%
Pharma	0%	0.648	-0.0061	-2.1	16.2%	0.00%
Pharma	5%	0.670	+0.0161	5.5	16.5%	1.24%
Pharma	10%	0.648	-0.0061	-2.1	17.6%	0.00%
Pharma	15%	0.655	+0.0009	0.3	19.6%	0.00%
Pharma	20%	0.655	+0.0015	0.5	19.8%	0.00%
Pharma	25%	0.656	+0.0019	0.7	22.1%	0.00%
Pharma	30%	0.676	+0.0219	7.5	25.7%	1.92%
Pharma	Full set-aside	0.948	+0.2941	100.0	100.0%	41.99%

Bayes–Nash simulation with $B = 3,000$ Monte Carlo draws per cell. Preference rate k scores SME bids by a factor $1 - k$ for winner selection while paying the actual winning bid. Δ is the simulated price effect relative to S_1 , in reference-price units. Welfare loss is $(DWL^{alloc} + \lambda \Delta G)/p_{S_1}$ at $\lambda = 0.30$. Entries that are negative before rounding are reported as 0.00%, since they lie inside Monte Carlo noise around a near-zero welfare effect.

Table OA-13: Set-aside versus preference ranking across λ

λ	Non-pharma		Pharma		
	Loss(V0)	Ranking	Loss(V0)	Ranking, main	Ranking, strict inv.
0.15	24.0%	V3 > V0	38.4%	V3 > V0	V0 > V3
0.20	25.6%	V3 > V0	40.8%	V3 > V0	V0 > V3
0.30	28.7%	V3 > V0	47.0%	V3 > V0	V0 > V3
0.40	31.8%	V3 > V0	50.2%	V3 > V0	V0 > V3
0.45	33.3%	V3 > V0	52.5%	V3 > V0	V0 > V3

V0 is the full SME-only set-aside and V3 is the 10 percent preference benchmark. The non-pharmaceutical ranking is stable across the λ grid. In pharmaceuticals, the ranking depends on how the post-policy SME pool is modeled, not mainly on the marginal cost of public funds.

E.3 Annual scaling

Annual scaling is kept out of the main text because it is not part of the core identification exercise. For reference, Group 65 annual reference outlays are R\$345 million in non-pharmaceuticals and R\$363 million in pharmaceuticals. Applying the structural welfare losses and adherence rates between 30 and 70 percent gives an annual welfare-cost range of R\$38–89 million, or US\$11–25 million at an exchange rate of 3.50 R\$/US\$. This scaling is useful for magnitude, but it should not be read as a claim about all procurement categories or about dynamic long-run SME capacity.

References for the Online Appendix

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